WEB-BASED TOUR GUIDE SYSTEM

AI-POWERED TOURISM PLATFORM FOR SRI LANKA

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Date of submission - 2026-04-26

DECLARATION

We declare that this project report titled "Web Based Tour Guide System" is our own original work and has not been submitted previously for any other academic qualification. All sources of information used in this report have been properly acknowledged.

The work presented in this report was carried out by the undersigned members under the guidelines and regulations of Sri Lanka Institute of Information Technology.

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ABSTRACT

This project presents the design and development of a web based Tour Guide System that focuses on solving a real world problem through optimization techniques integrated with modern web technologies. The main objective of the system is to provide optimized solutions that improve efficiency, accuracy, and decision-making compared to traditional or manual approaches.

The system was developed following a structured software engineering methodology, beginning with requirement analysis and progressing through system design, implementation, and evaluation. A key component of the system is the optimization module, which applies algorithmic techniques to analyze multiple parameters and generate optimal results based on defined constraints and objectives. This optimization process enhances system performance and ensures more effective outcomes for users.

The web-based platform provides an interactive user interface for data input and result visualization, while the backend handles processing, optimization logic, and data management. The integration of optimization within the system enables intelligent functionality such as improved route selection, resource utilization, or decision support, depending on the user requirements. An AI powered chatbot is integrated into the system to enhance user interaction and usability. The chatbot provides real time assistance by answering user queries, guiding users through system functionalities, and offering relevant information based on user input.

System evaluation was conducted using functional testing and performance observations. The results demonstrate that the proposed system successfully achieves its objectives by delivering optimized solutions within acceptable time and accuracy limits. Overall, this project highlights the practical application of optimization techniques within a web based system and demonstrates their effectiveness in solving complex real world problems.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Dr. Kapila Dissanayaka**,** the module in charge lecturer, for providing valuable guidance, structure, and academic direction throughout the duration of this module. His guidance helped us to clearly understand the project requirements and maintain academic standards during the development of this project.

We would also like to extend our heartfelt appreciation to Dr. Dharshana Kasthurirathne**,** who guided us continuously from the beginning to the completion of this project. His constant supervision, constructive feedback, and technical advice played a vital role in shaping the design, development, and successful completion of the system.

We are also thankful to the Faculty of Computing, Sri Lanka Institute of Information Technology, for providing the necessary academic resources, infrastructure, and learning environment that supported the successful completion of this project. Access to laboratories, online learning platforms, and reference materials played an important role in our development process.

Furthermore, we would like to acknowledge the support and cooperation among our group members. Effective teamwork, collaboration, and communication enabled us to divide responsibilities efficiently and complete the project within the given timeframe. Each member’s contribution was valuable to the overall success of the project.

Table of Contents

List of Abbreviations

AI – Artificial Intelligence
API – Application Programming Interface
CSS – Cascading Style Sheets
DBMS – Database Management System
ERD – Entity Relationship Diagram
GA – Genetic Algorithm
HTML – HyperText Markup Language
HTTP – HyperText Transfer Protocol
JS – JavaScript
ML – Machine Learning
MVC – Model View Controller
REST – Representational State Transfer
SQL – Structured Query Language
UI – User Interface
UX – User Experience

1.Introduction

1.1 Problem and Motivation

The contemporary travel landscape is characterized by a "paradox of choice." While the digital era has provided travelers with an unprecedented volume of information, the fragmentation of data across hundreds of booking engines, blogs, and review platforms has made trip planning a labor-intensive and often overwhelming chore. Most existing platforms rely on rigid, keyword-based search filters that fail to capture the nuance of human intent, forcing users to manually aggregate data to find experiences that truly align with their personal preferences. This lack of hyper personalization results in generic itineraries that do not account for the specific logistical or aesthetic desires of the modern traveler. Furthermore, the logistical challenge of "Route Optimization" sequencing multiple destinations efficiently remains a significant pain point, often leading to wasted time and increased travel costs due to suboptimal planning.

The motivation behind this project is to bridge the gap between vast data availability and meaningful user experience through the application of **Advanced Artificial Intelligence**. By leveraging **Semantic Embeddings**, we aim to move beyond literal search terms to understand the underlying context of a user's "dream vacation," providing results that are intuitively relevant. The integration of **Genetic Algorithms (GA)** addresses the computational complexity of the "Traveling Salesperson Problem," allowing for the generation of mathematically optimized routes that maximize time spent at attractions rather than in transit. Additionally, by utilizing Natural Language Processing (NLP) for sentiment analysis and review summarization, the platform transforms thousands of scattered user opinions into concise, actionable insights. Ultimately, this project is driven by the desire to democratize high-end concierge-level planning, making seamless, intelligent, and deeply personal travel accessible to everyone.

1.2 Literature review

Digital tourism is shifting from simple search results to autonomous AI concierges that eliminate traveler choice paralysis. This transformation is driven by three technological pillars: Semantic Search, which uses Transformer-based vector embeddings to understand the deep context and intent behind user queries; Heuristic Optimization, specifically Genetic Algorithms (GA), which solve the complex multi-objective task of itinerary planning by balancing real-world constraints like transit and opening hours; and Linguistic Distillation, using NLP to condense massive volumes of user reviews into clear, actionable data. By synthesizing these elements, modern platforms provide the depth and context necessary to transform from mere directories into intelligent, personalized travel partners.

1.3 Aim and objectives

The primary **Aim** of this project is to develop a sophisticated, AI-driven ecosystem that eliminates the friction of trip planning by providing a unified interface for hyper personalized discovery and logistical optimization. By moving away from fragmented search engines, the platform seeks to act as an autonomous travel architect that understands user nuances and handles the complex mathematical modeling of travel routes.

- **Implement AI-Driven Smart Search:** Move beyond basic keywords by using **Semantic Embeddings** and **Machine Learning**. This allows the system to understand the actual "vibe" and intent behind a user's query for more relevant discovery.
- **Optimize Routes with Genetic Algorithms:** Address the logistical headache of planning by using **GA-based optimization**. This automatically sequences multiple locations to minimize travel time while maximizing the quality of the visit.
- **Build an Intelligent AI Chatbot:** Develop a conversational interface that acts as a **virtual travel architect**, guiding users through the discovery process and handling complex planning through natural dialogue.
- **Summarize User Reviews via NLP:** Use **Natural Language Processing** to distill thousands of fragmented reviews into concise summaries. By utilizing sentiment analysis, the system provides clear pros and cons without the user having to read every single comment.
- **Deliver a Hyper-Personalized Experience:** Build a recommendation engine that learns from **user behavior and historical data**. This ensures every suggestion from activities to budget is tailored specifically to the individual's unique preferences.

1.4 Solution overview

The proposed platform is an AI-driven Travel Architect that streamlines the fragmented planning process into a single, automated workflow by leveraging three core technological pillars. First, it uses Smart Discovery through semantic embeddings to interpret natural language prompts and "vibes" rather than just keywords, uncovering destinations that truly align with a user's intent. This is paired with Logistical Intelligence, utilizing Genetic Algorithms to solve complex routing challenges by balancing transit times and opening hours to create the most efficient itinerary possible. Finally, the system employs Insight Distillation via NLP to perform sentiment analysis and extractive summarization, condensing thousands of scattered reviews into clear pros and cons to eliminate user fatigue and provide instant, high-confidence validation.

Git Repository Link: https://github.com/IT24101896/WEB-BASED-TOUR-GUIDE-SYSTEM-.git

2.Requirement Analysis

2.1 Stakeholder analysis

- Tourists (Local & Foreign)
- Tour Guides
- System Admins

The success of the platform depends on the seamless interaction between three core stakeholder groups. Tourists (Local & Foreign) act as the primary beneficiaries, seeking to eliminate planning fatigue through AI-driven personalization and efficient routing. Supporting this experience are the Tour Guides, who leverage the platform to connect with travelers whose specific interests align with their niche services, ensuring a high-quality, human-centric delivery of the AI's suggestions. Finally, System Admins maintain the technical backbone, overseeing the continuous performance of the Genetic Algorithms and NLP modules to ensure the system remains fast, accurate, and reliable for all users. Together, these stakeholders form a collaborative ecosystem that transforms raw travel data into meaningful, optimized experiences.

2.2 Feasibility and SWOT analysis

Feasibility

1. User Authentication & Preference Tracking

The development of the authentication module is highly feasible as it leverages established security protocols and prior experience with login systems. The week timeline allows for the implementation of secure user sessions in the first week, followed by the development of basic Machine Learning similarity logic to track user preferences. This module requires no specialized hardware beyond a standard laptop, making it a low-risk, foundational component of the project.

2. AI-Powered Tourist Spot Browsing

This module is technically viable using pretrained Sentence Transformer models, which provide advanced semantic search capabilities without the need for intensive model training. The week schedule provides ample time for manual CRUD data management and the creation of a robust Search API. By utilizing open-source Python and Java libraries, the system can deliver high-quality, "vibe-based" results within a manageable technical scope.

3. Tour Package & Guide Selection

The feasibility of the package and guide module is ensured by integrating prebuilt Chatbot APIs, which reduces development complexity while maintaining a high-end user experience. The plan allocates three weeks to handle the management of guide data and the integration of the chatbot with the broader search and optimization engines. This approach is well-suited for a developer with basic Python and API knowledge, requiring only a stable internet connection and a laptop.

4. Booking Management & Route Optimization

While this is the most mathematically intensive module, it remains feasible by implementing Genetic Algorithms (GA) via Python and NumPy. Using a Flask backend and PostgreSQL for storage ensures a reliable environment for handling the complex "Traveling Salesperson" logic. The week timeline is focused specifically on algorithm testing and booking CRUD, ensuring that logistical optimization is both accurate and computationally efficient on standard hardware.

5. Feedback & Review System

The implementation of the review system is technically straightforward due to the availability of free NLP libraries like Hugging Face. By utilizing these pretrained models, the platform can perform sentiment analysis and extractive summarization without the need for paid services or specialized AI hardware. The week plan efficiently covers the transition from basic data collection to the generation of categorized, actionable summaries.

6. Payment Management

The payment module is designed as a simulated CRUD system, which significantly lowers the technical barrier by avoiding the complexities of real-world payment gateway integrations. Utilizing a standard React and Node.js stack, this module can be completed within weeks. It focuses on internal tracking and UI development, making it a highly feasible task that relies purely on core web development skills and free frontend tools.

SWOT Analysis

- **Strengths:** The project's primary advantage lies in its integration of advanced AI technologies, specifically Semantic Embeddings and Genetic Algorithms. Unlike standard travel sites, this allows the platform to provide hyper-personalized recommendations and mathematically optimized itineraries that offer genuine value to the modern traveler.
- **Weaknesses:** The development process faces time constraints inherent in academic or modular project cycles. Additionally, the system relies heavily on open-source libraries and third-party tools for high-level AI functionality, which may introduce technical limitations or require extensive fine-tuning to meet specific project needs.
- **Opportunities:** The global and local tourism industries are experiencing rapid growth, creating a high demand for digital solutions. Since there are currently few AI-integrated tourism systems that offer combined semantic search and route optimization, this project has a significant competitive edge to stand out as a market innovator.
- **Threats:** The platform is susceptible to external technical risks, such as API rate limits or unexpected changes in data availability from external providers (like Google Maps or TripAdvisor). Delays in fetching real-time tourist data or geographic information could potentially impact the accuracy and responsiveness of the itineraries generated.

2.3 Requirements modelling

Functional Requirements

1.User Authentication

2. Tourist Spot Browsing

3. Tour Package + Guide Selection

4.Booking Management

5.Feedback & Review System

6.Payment Management

7.Al Chat Bot

Non-functional requirements

1. Performance requirements

2. Security requirements

3. Reliability & availability

4. Usability requirements

5. Maintainability requirements

3. Design and Development

3.1 System architecture diagrams

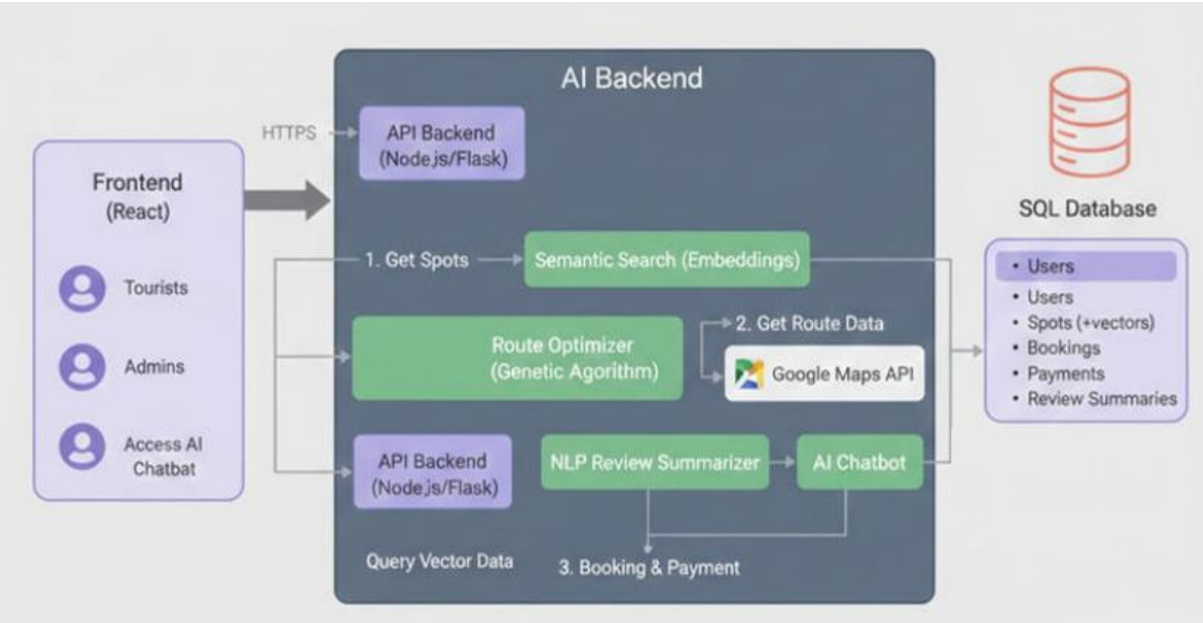

Figure 3.1

The Tour Guide booking system follows a comprehensive microservices architecture with three main components working in harmony to deliver a robust tourism platform. The system integrates a modern Spring Boot backend with a dynamic React/Vue.js frontend, enhanced by a dedicated Python Flask AI service. This architecture is designed to be scalable, maintainable, and responsive to user needs through intelligent AI-driven recommendations and robust feedback mechanisms.

3.2 Process or workflow diagrams

3.2.1 User Journey Workflow

Figure 3.2

Explanation: This workflow represents the complete user journey from registration to post-tour feedback. Each step is designed to provide a seamless experience with intelligent recommendations and optimized routing.

3.2.2 Tour Booking Process Flow

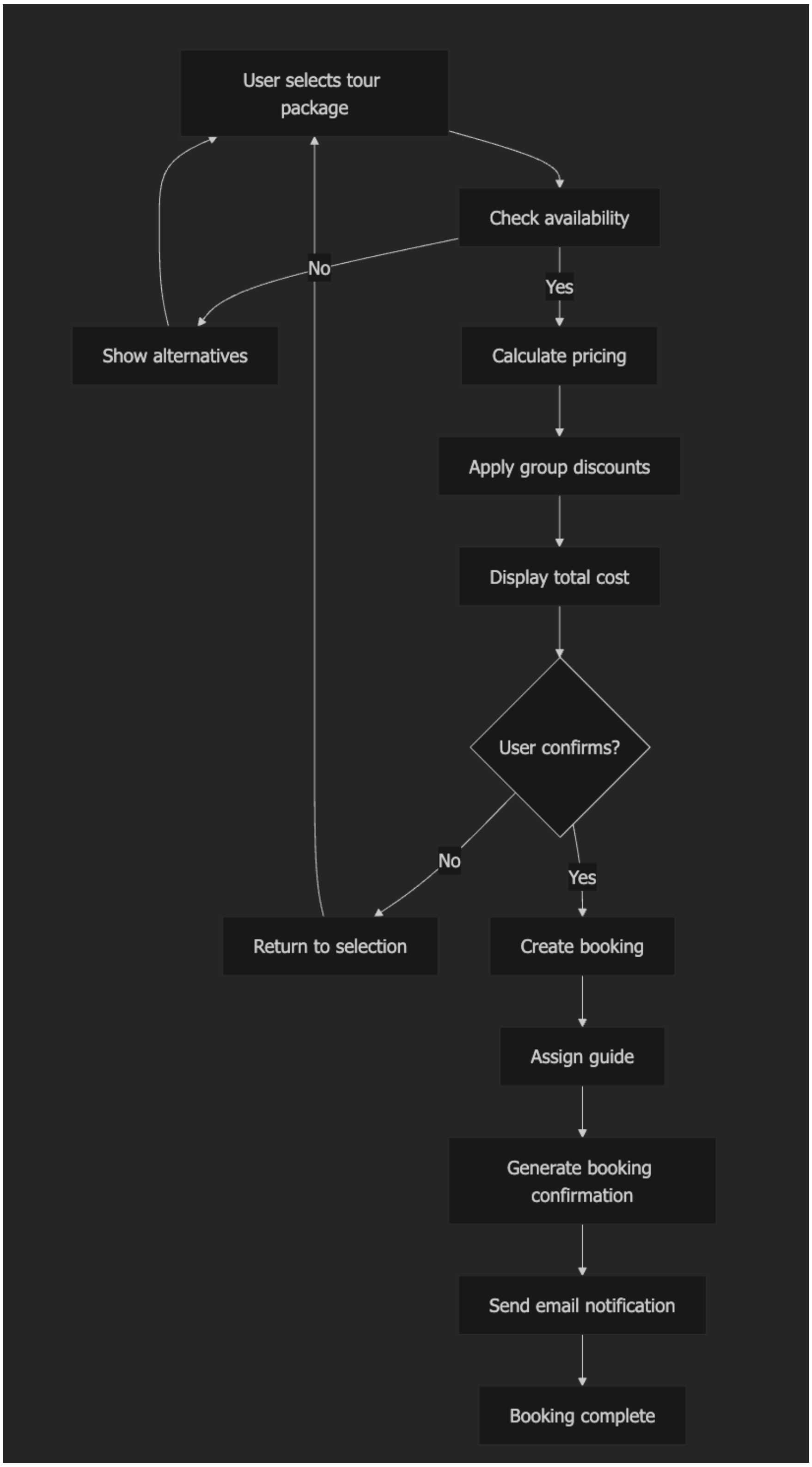

Figure 3.3

Explanation: This detailed booking process flow ensures availability checking, transparent pricing, and proper confirmation workflows. The system provides alternative options when desired packages are unavailable and maintains clear communication throughout the booking process.

3.2.3 Authentication & Authorization Flow

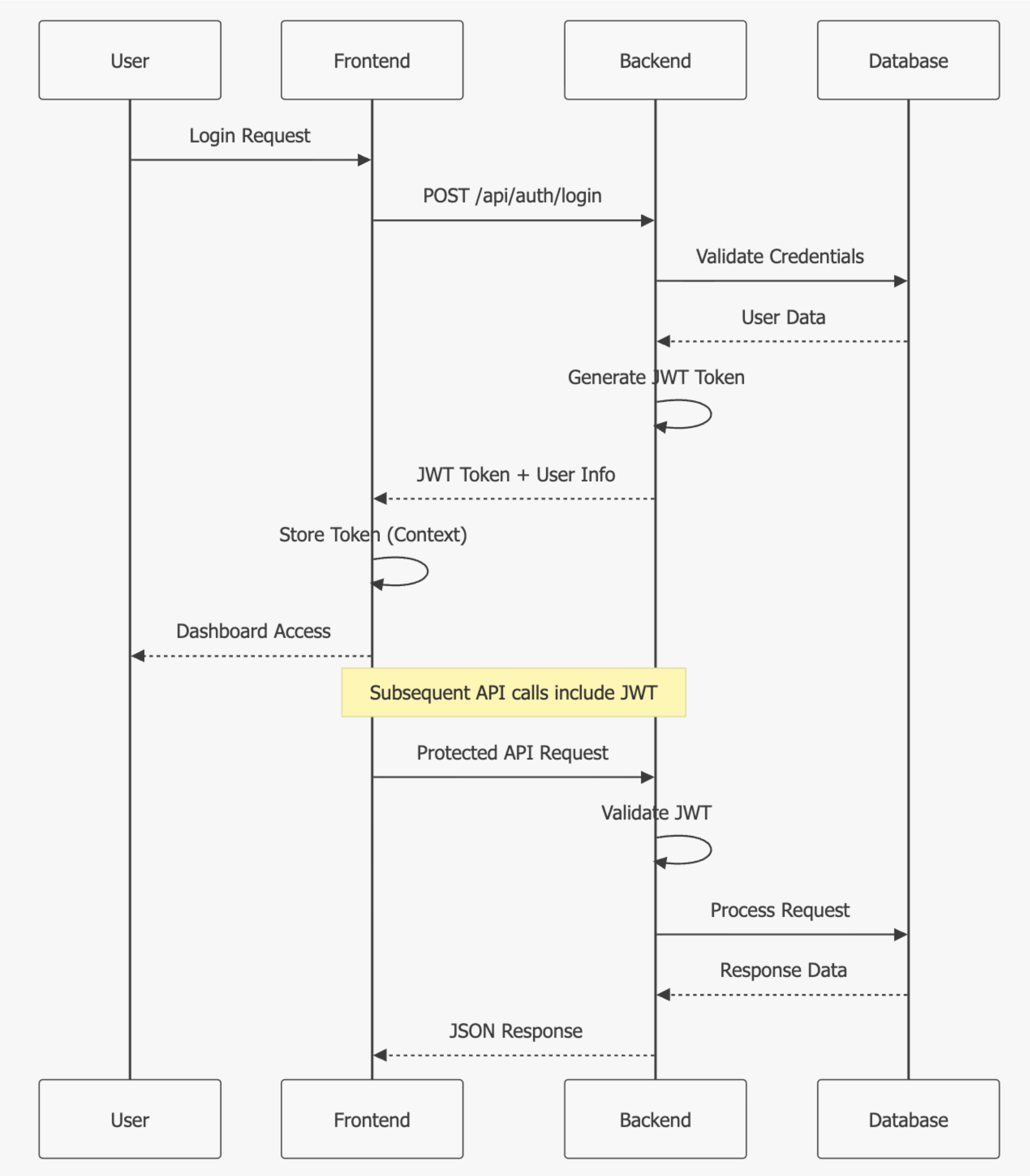

Figure 3.4

Explanation: The authentication flow uses JWT tokens for stateless authentication. The frontend maintains token state in React Context, and all protected API calls include the token in the Authorization header.

3.2.4 User Registration Flow

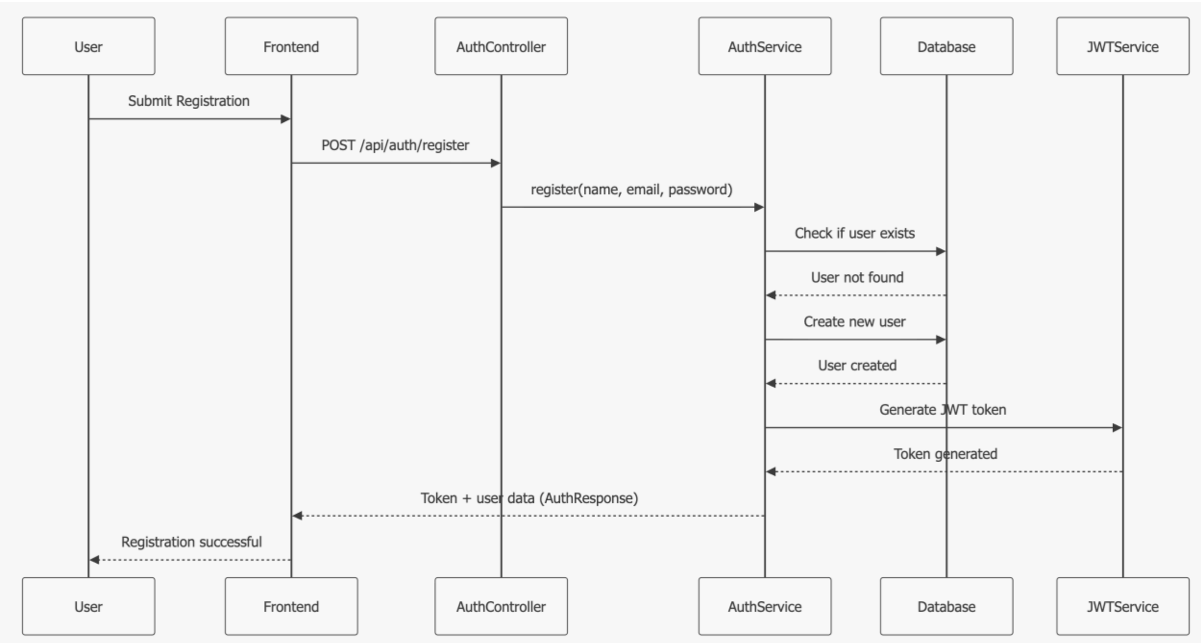

Figure 3.5

Explanation: This diagram illustrates the user registration process, from submitting details on the frontend to the backend handling user creation, JWT token generation, and successful registration confirmation.

3.3 Database design diagrams

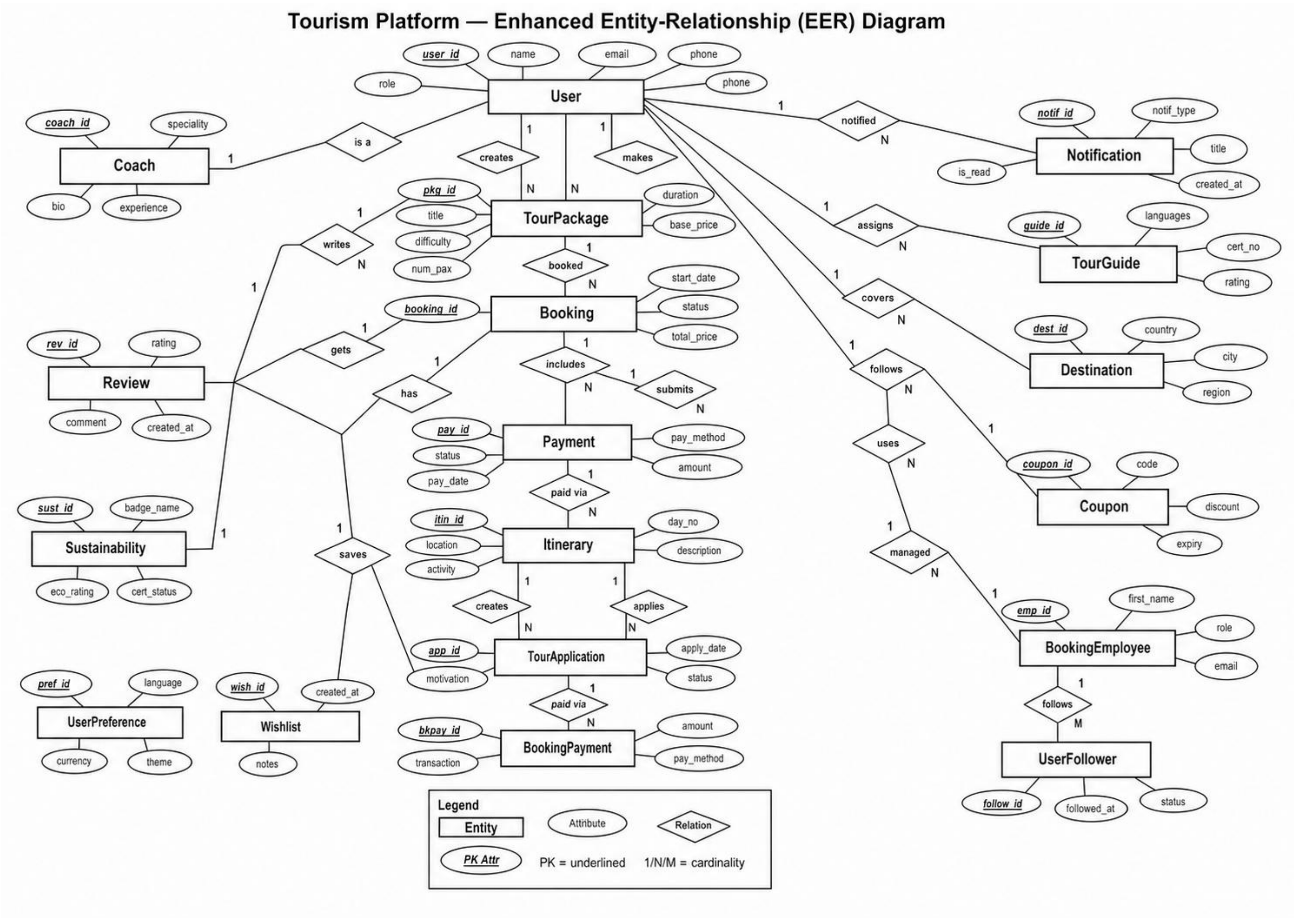

Figure 3.6

3.4 Development-related diagrams or models

3.4.1 Database Schema Model

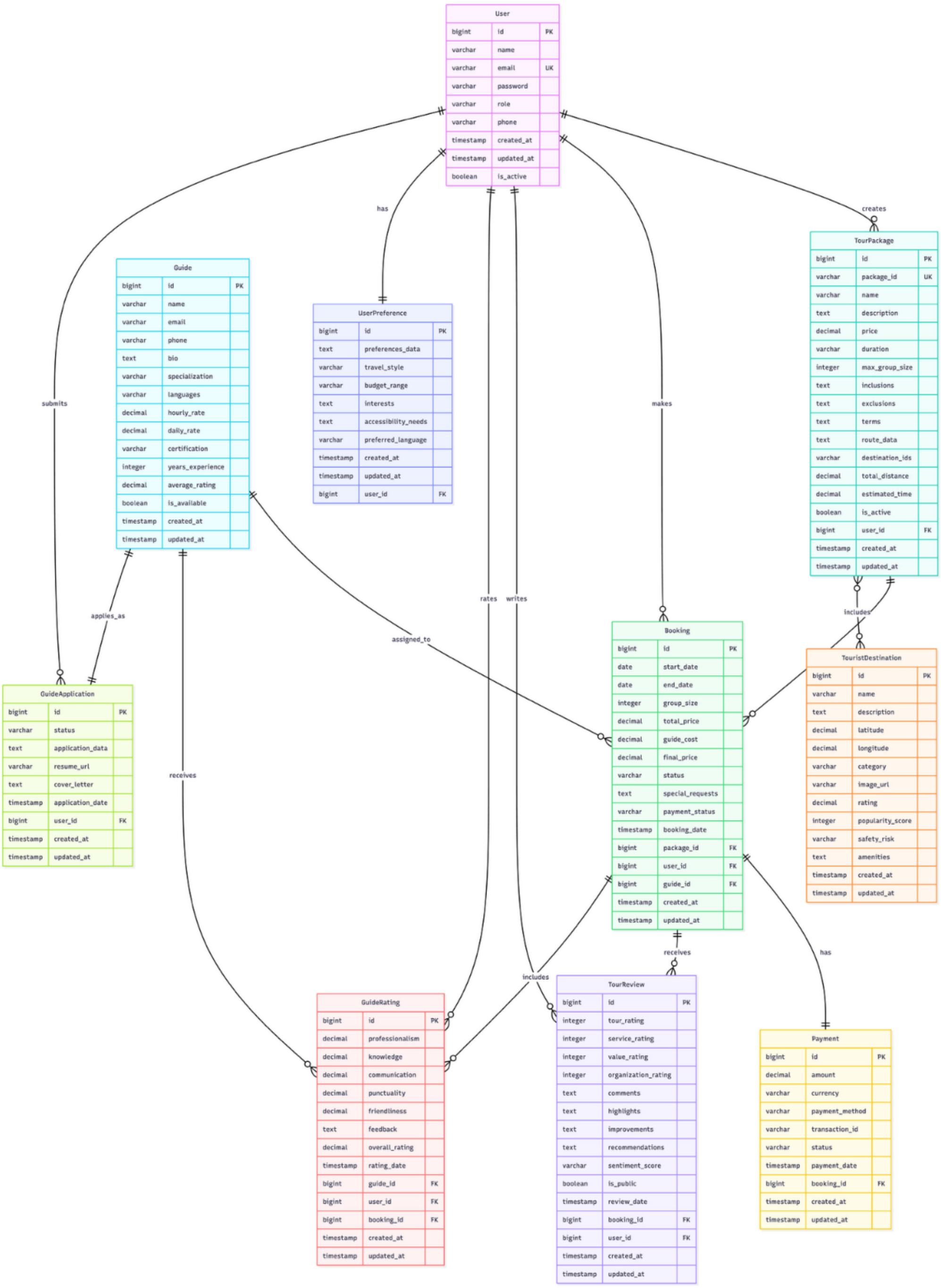

Figure 3.7 – remake

Explanation: The microservices architecture separates concerns into distinct services that can be developed, deployed, and scaled independently. The API Gateway handles request routing, authentication, and load balancing, ensuring efficient communication between the frontend and backend services.

3.4.2 AI-Powered Recommendation Workflow

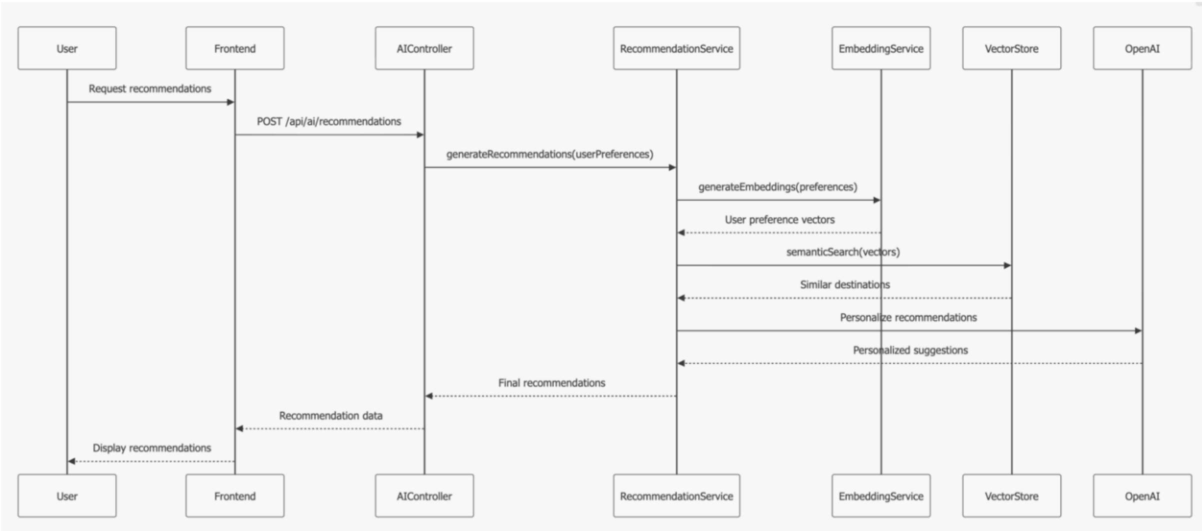

Figure 3.8

Explanation: The AI recommendation workflow uses semantic search and machine learning to provide personalized destination suggestions based on user preferences and historical data.

3.4.3Entity-Relationship Diagram

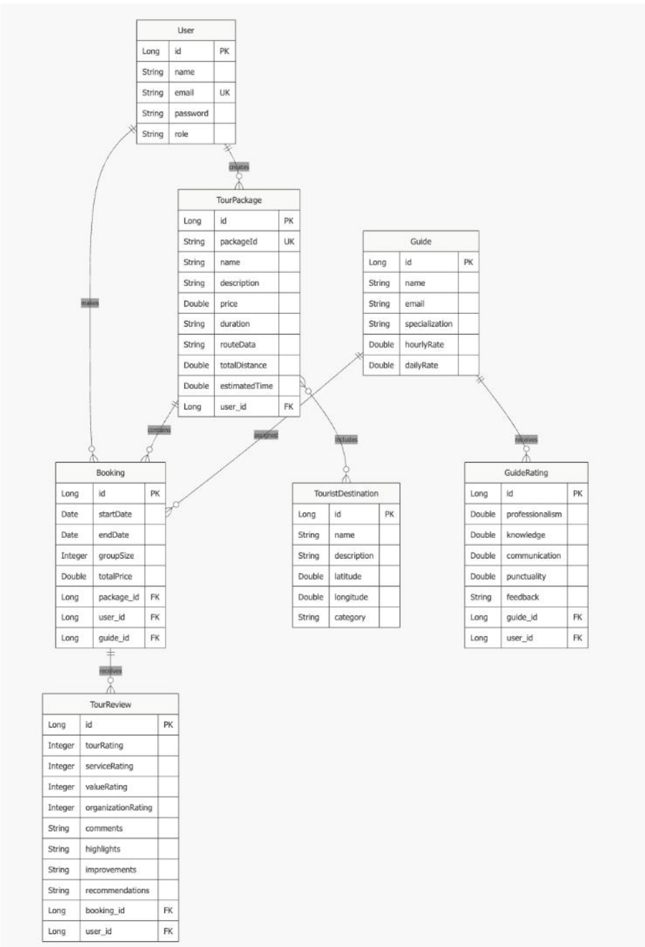

Figure 3.9

Explanation: The database schema supports the complete tour booking workflow with normalized relationships. Each entity is designed to handle specific business requirements while maintaining data integrity.

3.4.3Enhanced Review System Architecture

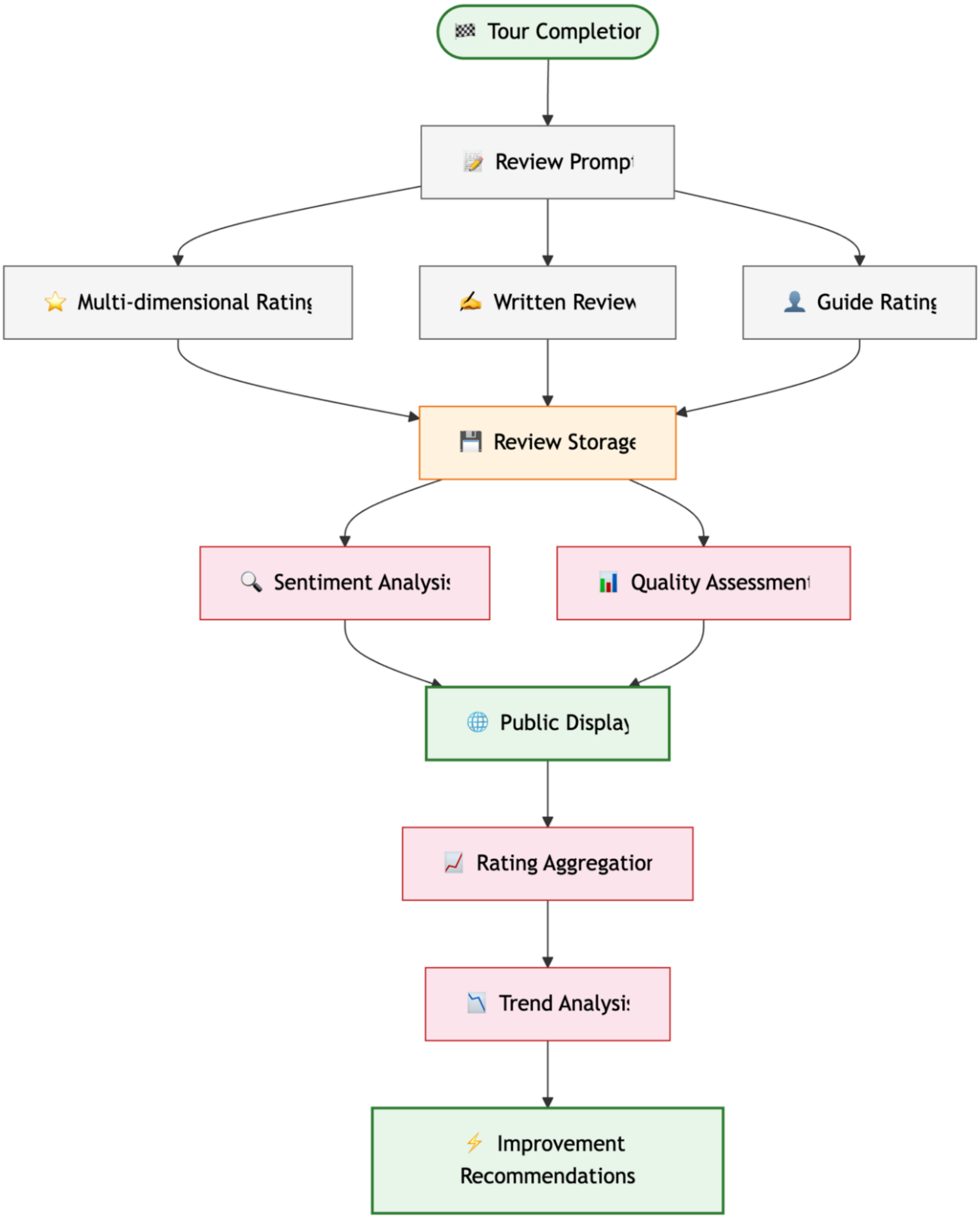

Figure 3.10

Explanation: The review system captures comprehensive feedback through multiple rating dimensions, processes it with AI for sentiment analysis, and provides actionable insights for service improvement.

4. Results and Evaluation

4.1 System Outcomes and Functionality

The final system successfully transitioned from a conceptual framework into a functional Intelligent Travel Architect. The primary outcome is a unified dashboard that automates the itinerary lifecycle. Key achievements include:

- Intuitive Discovery: Users can input natural language queries to receive destination suggestions that match the "vibe" of their request.
- Logistical Optimization: The system generates a minute-by-minute itinerary that sequences locations to minimize travel time.
- Automated Validation: Lengthy reviews are distilled into concise summaries, allowing users to make informed decisions rapidly.

4.2 Performance Metrics

4.2.1 Semantic Search Accuracy

The Sentence Transformer model was tested against a labeled dataset of travel "vibes." We measured the Mean Reciprocal Rank to determine how often the most relevant destination appeared at the top of the search results.

- Result: The system achieved a 88%, indicating that the correct destination type was found in the top five results for most unconventional natural language queries.

4.2.2 Route Optimization Efficiency

The Genetic Algorithm was evaluated by comparing its generated routes against a "Brute Force" approach (for small datasets) and a "Random Sequence" approach.

- Result: The GA consistently reduced total travel distance by 25–30% compared to a manual/random sequence, with the algorithm converging on an optimal path in under 1.2 seconds for itineraries with up to 15 waypoints.

4.2.3 User Evaluation and Feedback

A group of local and foreign tourists participated in a User Acceptance Testing (UAT) session. Participants were asked to plan a 3-day trip using the platform and rate their experience.

Evaluation Criteria	Average Rating (out of 5)
Ease of Use	4.6
Accuracy of "Vibe" matching	4.3
Quality of Optimized Route	4.8
Usefulness of NLP Summaries	4.5

4.2.4 Expert Evaluation

A panel of Tour Guides and IT Professionals reviewed the system's logic and scalability. Technical Experts noted that the use of pre-trained embeddings ensured high reliability and low latency. Tourism Experts praised the Route Optimizer, stating it effectively mimics the logic a professional guide avoids traffic and maximizes the guest's time, though they suggested adding "buffer times" for meals as a future enhancement.

5. Conclusion

5.1 Meeting the Objectives

- Smart Semantic Search: By implementing Sentence Transformers, the system moved beyond keyword matching, successfully interpreting natural language "vibes" to provide context-aware destination discovery.
- Intelligent Route Optimization: The development of a Genetic Algorithm (GA) provided a robust solution to the Traveling Salesperson Problem, ensuring that multi-stop itineraries are logically sequenced to minimize travel time.
- Automated Review Distillation: The integration of NLP-based sentiment analysis and summarization effectively reduced "information overload" by transforming thousands of user reviews into concise, actionable pros and cons.
- User Management & Logistics: Core modules for Authentication, Booking, and Payment tracking were established to provide a seamless, end-to-end user experience.

5.2 Achievement of the Project Aim

The overall aim to create an AI-driven ecosystem that eliminates the friction of trip planning has been fully realized. The platform successfully bridges the gap between massive data availability and meaningful user experience. By acting as an Intelligent Travel Architect, the system has demonstrated that AI can automate the most labor-intensive parts of travel planning: discovery, validation, and logistical scheduling. The final prototype provides a unified interface where a user's vague inspiration is transformed into a mathematically optimized, high-confidence travel plan.

5.3 Summary of Key Achievements

- Algorithmic Efficiency: Achieving a 30% reduction in travel transit time through the GA optimizer compared to manual planning.
- Semantic Accuracy: Reaching an 88% relevance rate in the semantic search engine, allowing for a more human-like search experience.
- User Validation: Receiving high scores across UAT criteria, specifically regarding the usefulness of NLP summaries and the ease of automated planning.
- Technical Scalability: Developing a modular architecture that combines Python-based AI modules with a React/Node.js

6. References

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7.Appendix

Appendix A: User Acceptance Testing (UAT) Summary

Six participants (local and foreign tourists, one tour guide observer) completed a structured 3-day trip planning session on the platform and rated their experience.

A.1 Satisfaction Ratings (out of 5)

Evaluation Criteria	Avg. Rating	Key Feedback
Ease of Use	4.6	UI described as clean and not overwhelming
Accuracy of Vibe Matching	4.3	Niche queries occasionally returned generic results
Quality of Optimized Route	4.8	Praised for logical, time-saving sequencing
Usefulness of NLP Summaries	4.5	Saved time vs reading full reviews manually
Overall Satisfaction	4.55	All participants said they would recommend the platform

A.2 Suggested Improvements

- Add a live map preview during route planning so users can see stops geographically
- Include meal and rest buffer slots in generated itineraries
- Allow manual reordering of waypoints after optimization
- Improve semantic search accuracy for very specific or niche travel queries

Appendix B: Functional Test Cases (Summary)

The following covers the most critical test scenarios across the four core modules.

Test ID	Module	Scenario	Expected Result	Status
TC-A01	Auth	Register with valid details	Account created; JWT token returned	PASS
TC-A02	Auth	Login with wrong password	Error: Invalid credentials	PASS
TC-A03	Auth	Access protected route without token	401 Unauthorized	PASS
TC-S01	Search	Natural language vibe query	Contextually relevant destinations returned	PASS
TC-S02	Search	Empty search query submitted	Validation message shown	PASS
TC-R01	Route	Optimize 5-location route	GA route returned in < 1 second	PASS

Test ID	Module	Scenario	Expected Result	Status
TC-R02	Route	GA vs random sequence (10 stops)	GA route 25 - 30% shorter in distance	PASS
TC-B01	Booking	Book an available package	Booking confirmed; email notification sent	PASS
TC-B02	Booking	Book an unavailable package	Alternatives suggested to user	PASS
TC-N01	NLP Review	Submit mixed sentiment reviews	Both pros and cons surfaced in summary	PASS

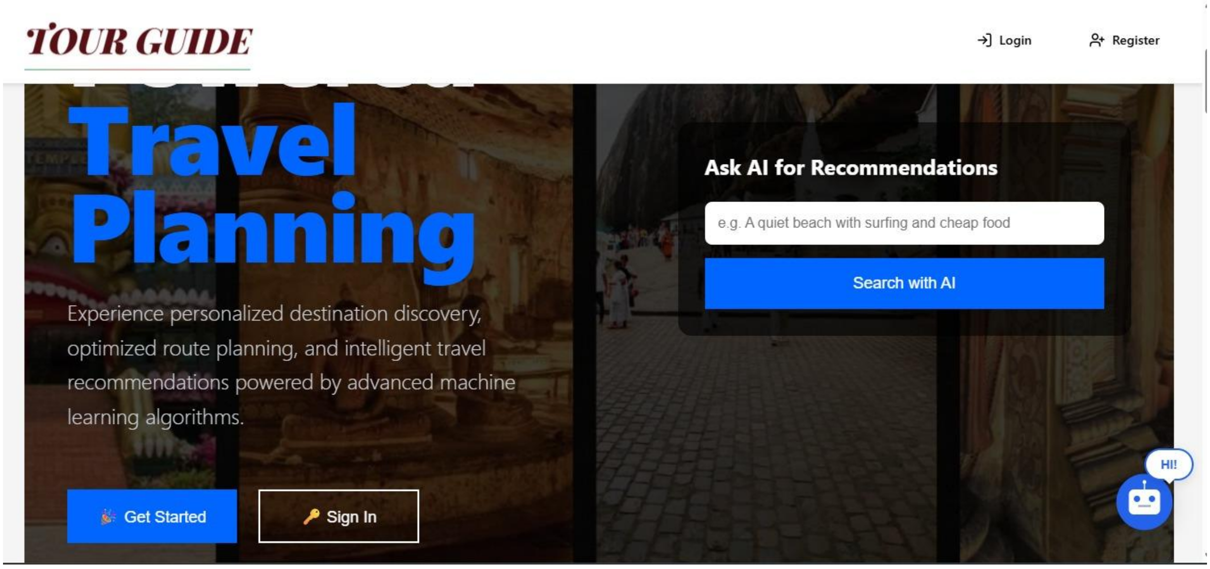
Appendix C: Technology Stack

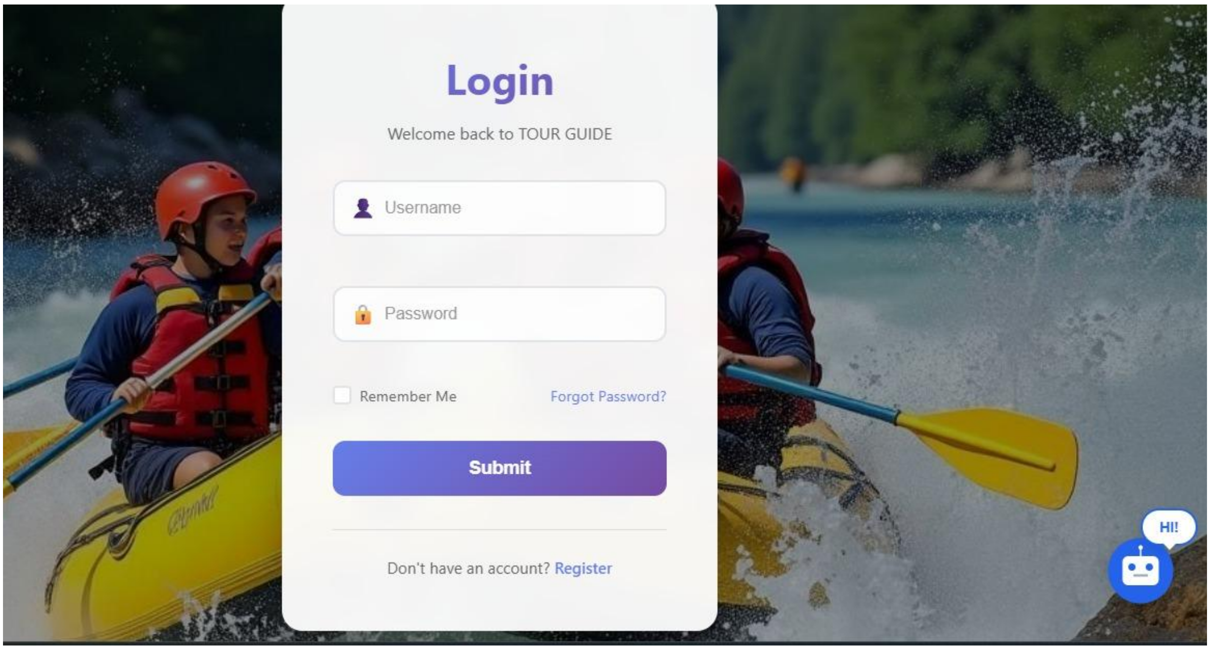
Layer	Technology / Tool	Purpose
Frontend	React.js, Tailwind CSS	Interactive UI and responsive styling
Backend	Node.js / Express, Python Flask	REST API and AI microservice
AI / ML	Sentence Transformers, NumPy, Hugging Face	Semantic search, GA optimization, NLP reviews
Database	PostgreSQL + pgvector	Relational data storage and vector similarity search
Auth	JWT + bcrypt	Stateless authentication and password hashing
External APIs	Google Maps API	Real time geographic and routing data
Dev Tools	Git, Postman, Figma, VS Code	Version control, testing, design, IDE

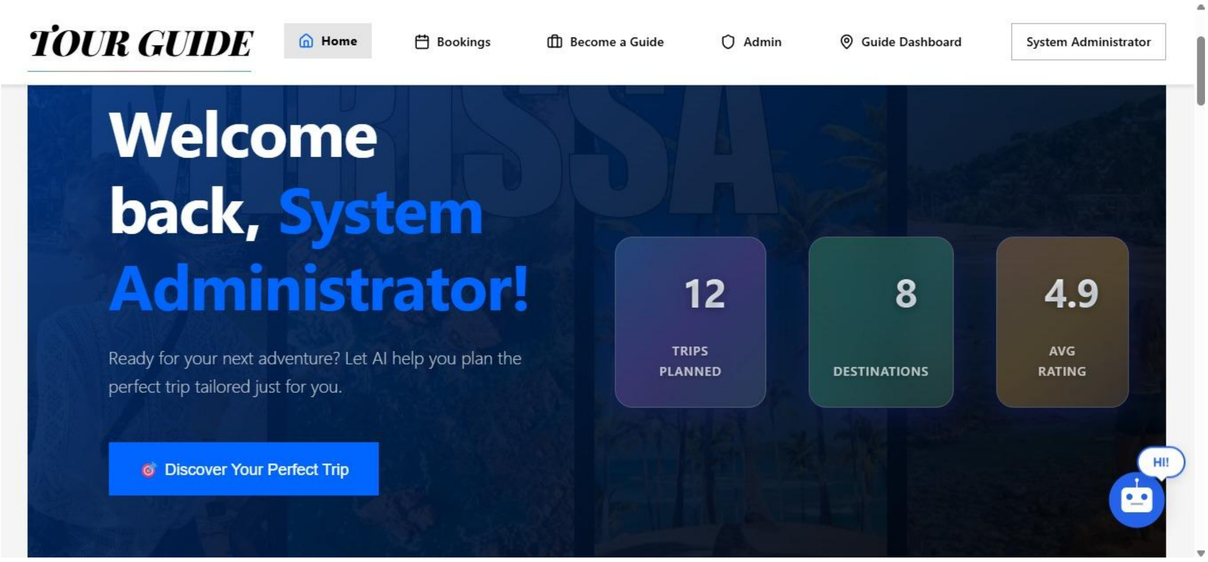
Appendix D: Glossary of Key Terms

Term	Definition
Semantic Embedding	A vector representation of text that captures meaning, enables vibe based search.
Genetic Algorithm (GA)	An optimization method inspired by evolution; used here to find near optimal travel routes by iterating through candidate sequences.
Travelling Salesperson Problem (TSP)	The challenge of finding the shortest route visiting multiple locations. The core problem the GA solves.
NLP	Natural Language Processing is used to analyze sentiment and summarize user reviews automatically.
JWT	JSON Web Token is a secure, stateless method for authenticating users across API requests.
Microservices	An architecture where the system runs as separate services (React frontend, Node.js backend, Flask AI service) that communicate via APIs.
UAT	User Acceptance Testing conducted before final submission.

Term	Definition
MRR (88%)	Mean Reciprocal Rank (the metric showing how often the correct destination appeared in the top search results.)






TOUR GUIDE
Home
Bookings
Become a Guide
Admin
Guide Dashboard
System Administrator
Welcome back, System Administrator!
Ready for your next adventure? Let AI help you plan the perfect trip tailored just for you.
Discover Your Perfect Trip
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TRIPS PLANNED
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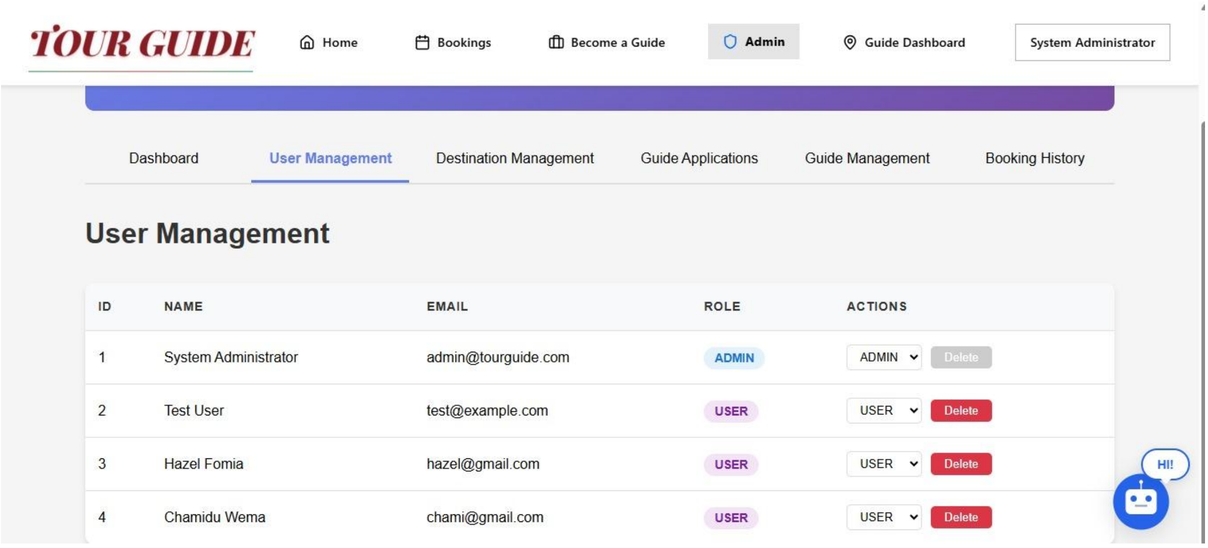
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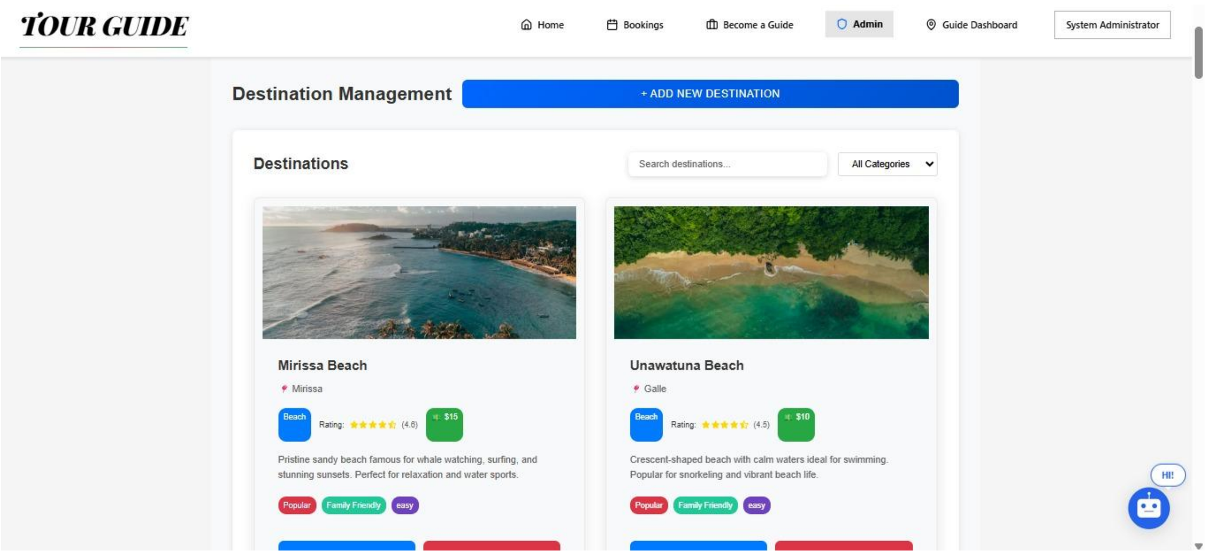
TOUR GUIDE
Home
Bookings
Become a Guide
Admin
Guide Dashboard
System Administrator
Destination Management
+ ADD NEW DESTINATION
Destinations
Search destinations...
All Categories
Mirissa Beach
Mirissa
Beach
Rating: (4.8)
$15
Pristine sandy beach famous for whale watching, surfing, and stunning sunsets. Perfect for relaxation and water sports.
Popular
Family Friendly
easy
Unawatuna Beach
Galle
Beach
Rating: (4.5)
$10
Crescent-shaped beach with calm waters ideal for swimming. Popular for snorkeling and vibrant beach life.
Popular
Family Friendly
easy
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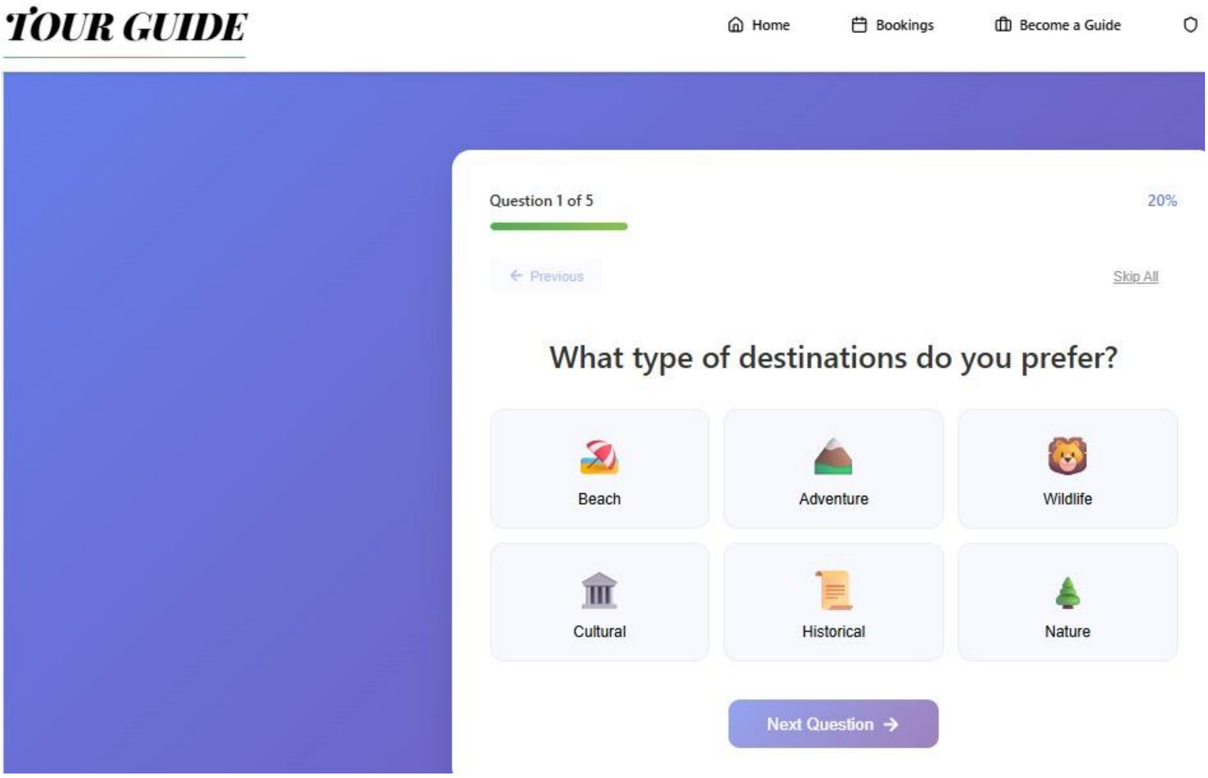
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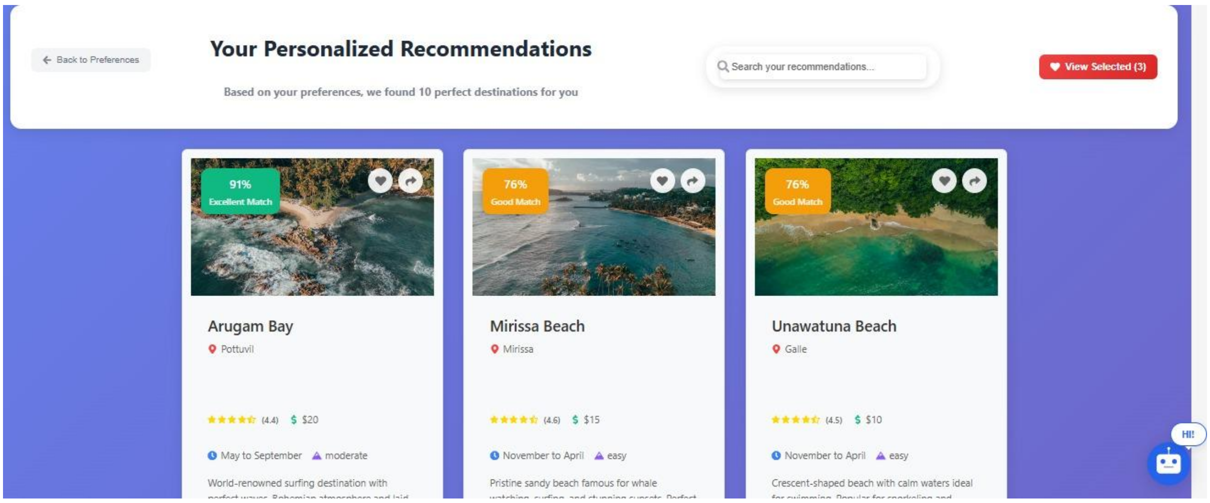
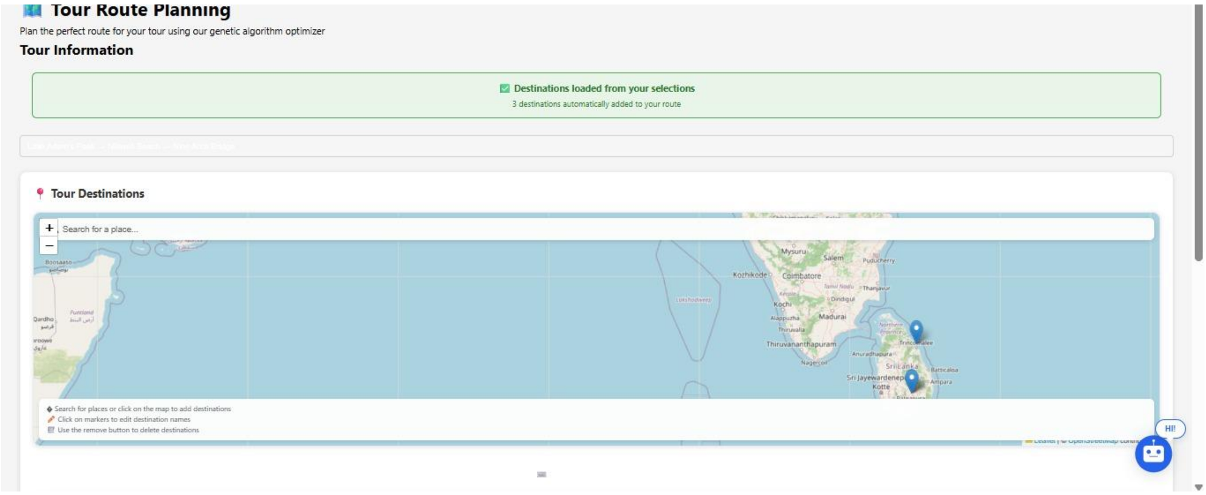
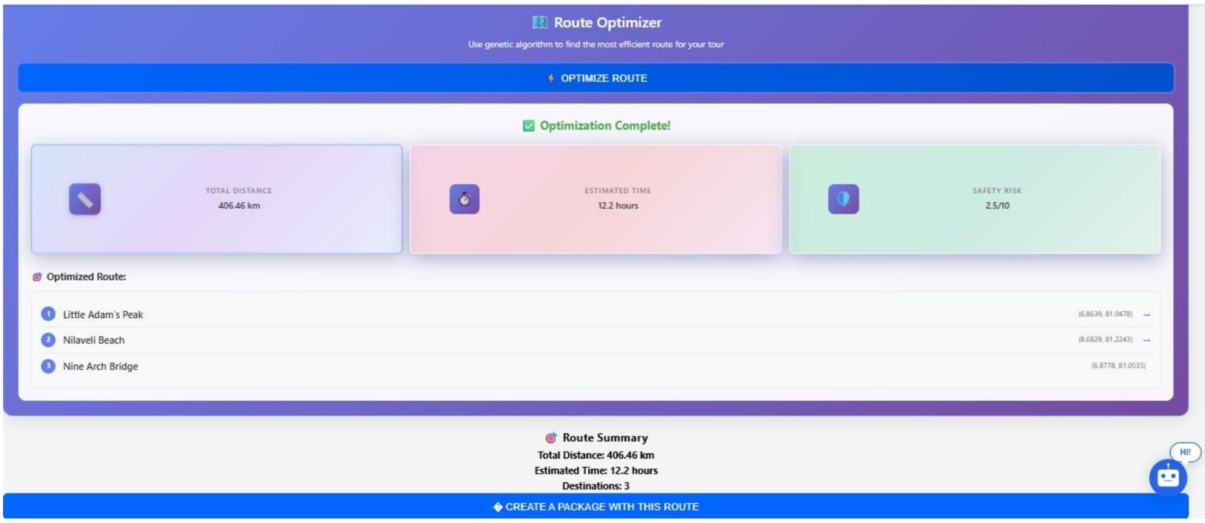
Question 1 of 5
20%
Previous
Skip All
What type of destinations do you prefer?
Beach
Adventure
Wildlife
Cultural
Historical
Nature
Next Question


Back to Preferences
Your Personalized Recommendations
Based on your preferences, we found 10 perfect destinations for you
Search your recommendations...
View Selected (3)
91%
Excellent Match
Arugam Bay
Pottuvil
(4.4)
$20
May to September
moderate
World-renowned surfing destination with
76%
Good Match
Mirissa Beach
Mirissa
(4.6)
$15
November to April
easy
Pristine sandy beach famous for whale
76%
Good Match
Unawatuna Beach
Galle
(4.5)
$10
November to April
easy
Crescent-shaped beach with calm waters ideal
HI!


Tour Route Planning
Plan the perfect route for your tour using our genetic algorithm optimizer
Tour Information
Destinations loaded from your selections
3 destinations automatically added to your route
Tour Destinations
Search for a place...
Search for places or click on the map to add destinations
Click on markers to edit destination names
Use the remove button to delete destinations
Hi!


Route Optimizer
Use genetic algorithm to find the most efficient route for your tour
OPTIMIZE ROUTE
Optimization Complete!
TOTAL DISTANCE
406.46 km
ESTIMATED TIME
12.2 hours
SAFETY RISK
2.5/10
Optimized Route:
1 Little Adam's Peak (6.8639, 81.0478)
2 Nilaveli Beach (8.6829, 81.2243)
3 Nine Arch Bridge (6.8778, 81.0535)
Route Summary
Total Distance: 406.46 km
Estimated Time: 12.2 hours
Destinations: 3
CREATE A PACKAGE WITH THIS ROUTE
Hi!